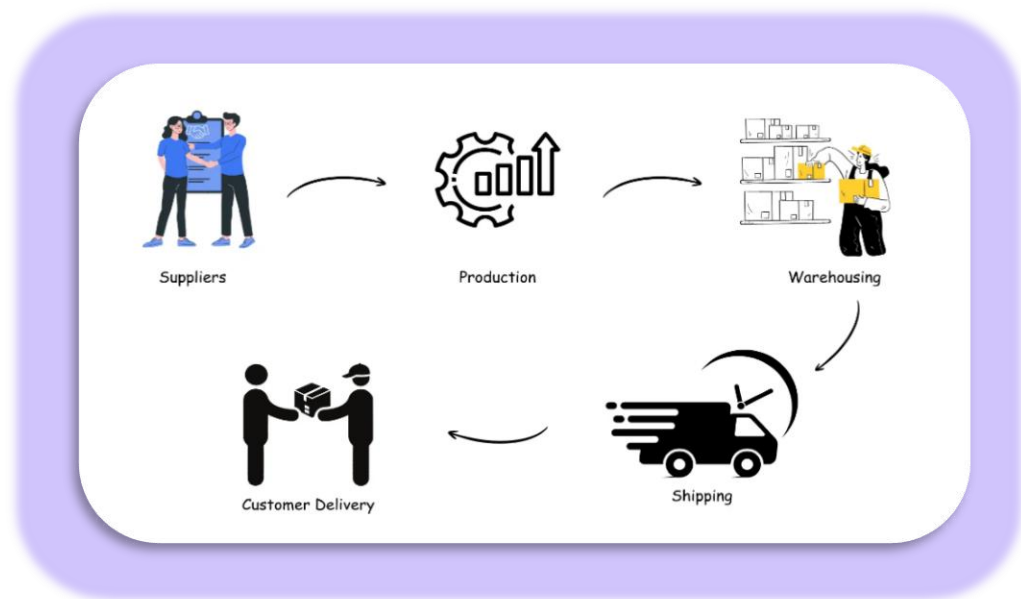


INTERNSHIP REPORT

Supply Chain Visibility Logistics



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ABSTRACT

Supply chain management involves coordinating inventory, suppliers, warehouses, transportation, and deliveries to ensure products reach customers efficiently. Raw operational data collected from these activities often contains missing values, duplicate records, inconsistent formats, and incorrect data types, making it unsuitable for direct analysis.

The objective of this project is to clean and preprocess a supply chain dataset so that it can be used for Exploratory Data Analysis (EDA), visualization, and Power BI dashboard development. The preprocessing process includes identifying missing values, removing duplicate records, standardizing data formats, correcting data types, and creating calculated fields required for analysis.

Python was used as the primary programming language, with Pandas and NumPy libraries for data preprocessing. Business logic was applied while handling missing values to preserve the consistency and reliability of the dataset.

After preprocessing, the dataset became structured, consistent, and ready for further analysis. The cleaned dataset will support dashboard development, performance monitoring, and data-driven decision-making for supply chain operations.

1. Introduction

Supply chain management is the process of managing the movement of products from suppliers to customers. It includes inventory management, warehouse operations, transportation, and product delivery. Every stage of the supply chain generates data that helps organizations monitor performance and improve operational efficiency.



Supply chain management

Raw supply chain data often contains missing values, duplicate records, inconsistent formats, and incorrect data types. These issues reduce data quality and may lead to inaccurate analysis and business decisions. Therefore, data preprocessing is an essential step before performing analysis or creating dashboards.

This project focuses on cleaning and preparing a supply chain dataset for further analysis. Using Python and its data analysis libraries, the dataset was preprocessed by handling missing values, validating data types, standardizing data, and creating calculated fields. The processed dataset will be used for Exploratory Data Analysis (EDA) and the development of an interactive Power BI dashboard to improve supply chain visibility.

2. Problem Statement

Supply chain organizations generate a large amount of operational data every day from inventory, suppliers, warehouses, transportation, and delivery activities. This data is useful for monitoring business performance and making operational decisions. However, raw datasets are often not ready for analysis because they contain missing values, duplicate records, inconsistent formats, and incorrect data types.

Using such data directly can produce inaccurate analysis, misleading visualizations, and unreliable business reports. For example, missing delivery dates or inconsistent transportation records can affect the calculation of delivery performance and other key performance indicators (KPIs).

To overcome these challenges, the dataset must be cleaned and preprocessed before it is used for analysis or dashboard development. Data preprocessing improves the quality, consistency, and reliability of the dataset by handling missing values, removing duplicate records, correcting data formats, and validating important fields.

This project focuses on preparing a clean and structured supply chain dataset that can be used for Exploratory Data Analysis (EDA), Power BI dashboard development, and data-driven decision-making.

3. Project Aim

The main aim of this project is to prepare the raw supply chain dataset for analysis by performing data cleaning and preprocessing. Raw data collected from supply chain operations may contain missing values, duplicate records, inconsistent formats, and incorrect data types. These issues can reduce the accuracy and reliability of data analysis.

The project focuses on improving the quality and consistency of the dataset by applying appropriate preprocessing techniques. A clean and well-structured dataset is essential for generating meaningful insights, creating effective visualizations, and supporting business analysis.

After preprocessing, the dataset becomes suitable for Exploratory Data Analysis (EDA), dashboard development, and future machine learning applications. The overall goal is to improve supply chain visibility and support better business decisions through accurate and reliable data.

4. Project Objectives

The main objectives of this project are:

- Understand the structure of the supply chain dataset.
- Clean and preprocess the raw dataset.
- Handle missing values and duplicate records.
- Standardize data formats and data types.
- Prepare the dataset for Exploratory Data Analysis (EDA) and visualization.
- Analyze delivery performance and transportation costs.
- Improve supply chain visibility.
- Support data-driven business decisions.

5. Dataset Description

- The dataset used in this project represents supply chain and logistics operations. It contains information related to customer orders, suppliers, warehouses, shipping methods, delivery schedules, transportation costs, and order values. The dataset is used to monitor different stages of the supply chain and evaluate delivery performance.
- Before preprocessing, the dataset was explored to understand its structure, identify missing values, and determine the required cleaning operations.

Dataset Information

Attribute	Details
Dataset Name	Supply Chain Logistics Dataset
Domain	Supply Chain and Logistics
Source	<i>To be Updated (Kaggle Dataset Link)</i>
File Format	Excel (.xlsx)
Number of Records	1000
Number of Columns	14
Data Types	Object, DateTime, Integer, Float

The dataset contains a combination of categorical, numerical, and date-related features, making it suitable for analyzing order processing, delivery performance, transportation .

- cost, and overall supply chain operations.

6. Dataset Features

The dataset contains **14 features**, each representing a specific stage of the supply chain process.

Column Name	Description
Order_ID	Unique identifier for each order.
Customer_ID	Unique identifier of the customer.
Product_ID	Unique identifier of the ordered product.
Supplier	Name of the supplier providing the product.
Warehouse	Warehouse from which the order is dispatched.
Order_Date	Date on which the customer placed the order.
Dispatch_Date	Date on which the order was dispatched.
Delivery_Date	Actual delivery date of the order.
Expected_Delivery_Date	Planned delivery date of the order.
Delivery_Status	Delivery status (Early, On Time, or Delayed).

Column Name	Description
Delay_Days	Difference between actual and expected delivery date in days.
Shipping_Mode	Transportation mode used for delivery (Road, Rail, Air, etc.).
Transportation_Cost	Cost incurred for transporting the order.
Order_Value	Total monetary value of the order.

Dataset Summary

The dataset contains:

- **1000 records**
- **14 columns**
- **7 Object columns**
- **4 DateTime columns**
- **2 Float columns**
- **1 Integer column**

During the initial exploration, missing values were identified in multiple columns, indicating the need for data preprocessing before analysis and visualization.

7. Tools and Technologies Used

The following tools and technologies were used to complete this project.

Tool / Technology	Purpose
Python	Programming language used for data preprocessing and analysis.
Pandas	Used to load, clean, manipulate, and preprocess the dataset.
NumPy	Used for numerical operations and calculations.
Jupyter Notebook	Development environment used to write and execute Python code.
Microsoft Excel	Used to store and review the dataset before and after preprocessing.
Power BI	Used for developing an interactive dashboard after preprocessing.

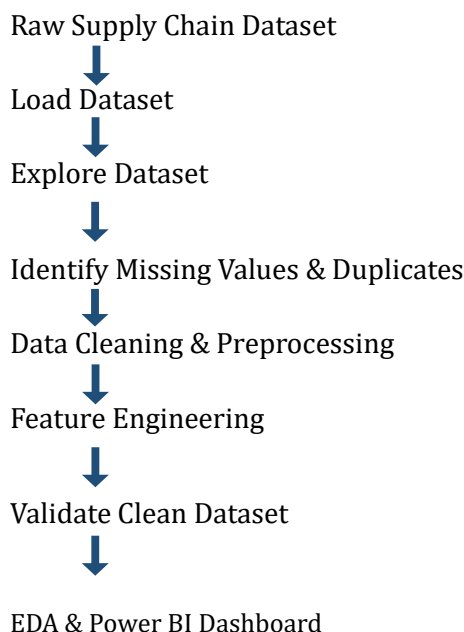
These tools provided an efficient workflow for data cleaning, preprocessing, analysis, and visualization.

8. Project Workflow

The project followed a systematic workflow to transform the raw supply chain dataset into a

clean dataset suitable for analysis and dashboard development.

Workflow



The workflow begins with loading and exploring the dataset. Data quality issues such as missing values and duplicate records are identified and corrected. After cleaning, new features are created where required, and the processed dataset is validated before being used for Exploratory Data Analysis (EDA) and Power BI dashboard development.

9. Data Preprocessing

Data preprocessing is the process of improving the quality of raw data before analysis. The supply chain dataset contained missing values and inconsistent information that needed to be corrected to ensure accurate analysis and reporting.

9.1 Import Libraries and Load Dataset

The required Python libraries were imported, and the dataset was loaded into a Pandas DataFrame.

Code

```
import pandas as pd
import numpy as np
df = pd.read_excel("Supply_Chain_Dataset.xlsx")
```

The dataset was successfully loaded into Python for preprocessing.

9.2 Dataset Exploration

The dataset was explored to understand its structure and identify potential data quality issues.

Functions Used

- `df.head()`
- `df.tail()`
- `df.shape`
- `df.columns`
- `df.info()`
- `df.describe()`

Observation

- Total Records: **1000**
- Total Columns: **14**
- Data Types: Object, DateTime, Float, and Integer
- Multiple columns contained missing values.

9.3 Data Quality Assessment

The dataset was checked for missing values and duplicate records.

Code

```
df.isnull().sum()
```

```
df.duplicated().sum()
```

Missing Values Summary

Column	Missing Values
Order_ID	3
Customer_ID	8
Product_ID	7
Supplier	8
Warehouse	8
Order_Date	8
Dispatch_Date	5
Expected_Delivery_Date	4
Delivery_Status	7
Shipping_Mode	2
Transportation_Cost	2

Column	Missing Values
Order_Value	1

The identified missing values were handled during the preprocessing stage.

9.4 Data Cleaning

The following preprocessing operations were performed:

- Filled missing values in identifier columns.
- Completed missing dates using the most common processing time.
- Filled categorical columns using suitable values.
- Filled numerical columns using appropriate statistical methods.
- Ensured consistency in data formats and values.

These steps improved the completeness and reliability of the dataset.

9.5 Feature Engineering

Two important features were prepared during preprocessing.

Delay_Days

The delivery delay was calculated by comparing the actual delivery date with the expected delivery date.

```
df["Delay_Days"] = (
    df["Delivery_Date"] - df["Expected_Delivery_Date"]
).dt.days
```

Delivery_Status

The delivery status was updated using the calculated delay.

```
df.loc[df["Delay_Days"] < 0, "Delivery_Status"] = "Early"
df.loc[df["Delay_Days"] == 0, "Delivery_Status"] = "On Time"
df.loc[df["Delay_Days"] > 0, "Delivery_Status"] = "Delayed"
```

This ensured that every order was correctly classified based on its delivery performance.

9.6 Final Validation

After preprocessing, the dataset was validated using:

```
df.info()
df.isnull().sum()
```

The cleaned dataset was then saved for further analysis.

```
df.to_excel("Cleaned_Supply_Chain_Dataset.xlsx", index=False)
```

The processed dataset is now ready for Exploratory Data Analysis (EDA) and Power BI dashboard development.

10. Exploratory Data Analysis (EDA)

After completing data preprocessing, Exploratory Data Analysis (EDA) was performed to understand the distribution of the data, identify trends, and discover meaningful patterns before developing the Power BI dashboard. The analysis focused on delivery performance, shipping methods, warehouse activities, transportation costs, and order values.

The following observations were identified during EDA:

- Delayed orders accounted for the largest share of deliveries, indicating that delivery performance requires improvement.
- Sea shipping was the most frequently used transportation mode among all shipping methods.
- The Pune warehouse processed the highest number of customer orders.
- FastTrack Distributors handled the highest number of product deliveries.
- Order value and transportation cost showed noticeable fluctuations over time, reflecting changes in business demand and logistics expenditure.

These observations provided the foundation for selecting Key Performance Indicators (KPIs) and designing the dashboard.

11. Key Performance Indicators (KPIs)

To evaluate the overall performance of the supply chain, several Key Performance Indicators (KPIs) were developed. These KPIs provide a quick summary of operational performance and help monitor business efficiency.

The dashboard includes the following KPIs:

- Total Orders
- Total Order Value
- Total Transportation Cost
- Average Delay Days
- On-Time Delivery Percentage

These KPIs enable users to monitor supply chain performance and quickly identify areas requiring improvement.

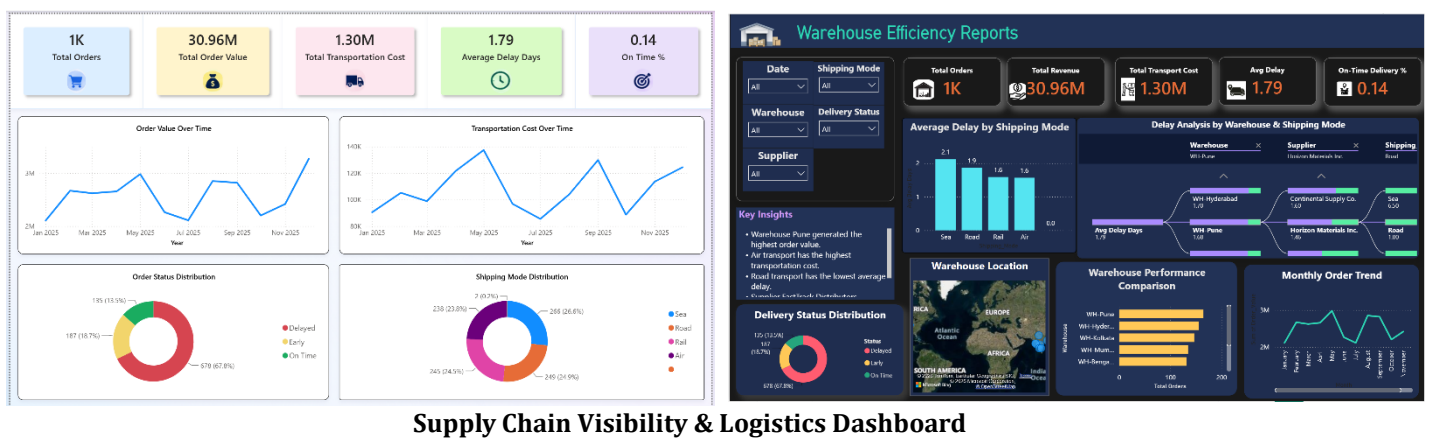
12. Power BI Dashboard Development

An interactive Power BI dashboard was developed using the cleaned dataset. The dashboard combines multiple visualizations into a single interface, allowing users to monitor warehouse performance, delivery status, shipping methods, and transportation costs.

The dashboard includes the following components:

- KPI cards providing an overall business summary.
- Charts comparing warehouse performance and shipping modes.
- Line charts showing order value and transportation cost trends over time.
- A map displaying warehouse locations.
- Interactive filters for warehouse, supplier, and delivery status.
- A decomposition tree for analyzing the causes of delivery delays.

The dashboard enables users to explore the data interactively and supports faster business decision-making.



13. Business Insights

Based on the dashboard analysis, the following business insights were identified:

- Most customer orders were delivered late, indicating opportunities to improve delivery performance.
- Sea transportation handled the highest number of shipments, making it the most commonly used shipping mode.
- Certain warehouses processed significantly more orders than others, indicating differences in warehouse workload.
- Transportation cost contributed substantially to overall logistics expenses.
- The percentage of on-time deliveries remained relatively low, highlighting the need for better delivery planning and operational efficiency.

These insights help organizations identify operational challenges and support data-driven decision-making.

14. Future Scope

The project can be further enhanced by implementing additional analytical and predictive capabilities. Possible future improvements include:

- Develop machine learning models to predict delivery delays.
- Integrate real-time logistics and shipment tracking data.
- Create executive-level dashboards for business management.
- Generate automated alerts for delayed shipments.
- Expand the dashboard by adding supplier and warehouse performance analytics.

10. Summary

This project focused on preparing a raw supply chain dataset for analysis by performing data preprocessing. The dataset was first explored to understand its structure, data types, and quality. Missing values and duplicate records were identified before applying appropriate preprocessing techniques.

During preprocessing, missing values were handled using suitable methods, data consistency was improved, and important features such as Delay_Days and Delivery_Status were prepared. The processed dataset was validated to ensure its accuracy and completeness.

The final cleaned dataset is now ready for Exploratory Data Analysis (EDA), Power BI dashboard development, and further business analysis.

11. Conclusion

The objective of this project was to improve the quality of the supply chain dataset through data preprocessing. By cleaning the dataset, handling missing values, validating data, and preparing new features, the dataset became more reliable and suitable for analysis.

The preprocessing phase provides a strong foundation for the next stages of the project, including Exploratory Data Analysis (EDA), dashboard development, and business reporting. A clean and well-structured dataset helps generate accurate insights, improve supply chain visibility, and support data-driven decision-making.

Overall, the project demonstrates the importance of data preprocessing as a key step in any data analytics workflow.

12. References

1. Kaggle Supply Chain Logistics Dataset (*Dataset link to be updated*)
2. Python Software Foundation. Python Programming Language. <https://www.python.org/>
3. Pandas Documentation. <https://pandas.pydata.org/docs/>

4. NumPy Documentation. <https://numpy.org/doc/>
5. Microsoft Power BI Documentation. <https://learn.microsoft.com/power-bi/>